Data **Analytics** on flight Delays



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BDDA-Workshop1-2026/2027

Problem, Objectives & Decisions

Problem

- Delays and cancellations are reported as one number, so it is hard to see when, where and why they happen.
- Passengers and airlines cannot tell which problems are avoidable.

Objectives

- Build a PostgreSQL database from a 50,044-flight sample of the 2015 BTS data.
- Analyse delays by airline, airport, time and cause, and analyse cancellations.
- Present the results in Streamlit dashboards and recommend improvements.

Decisions supported

- **Passengers:** which airlines, hours and months to avoid.
- **Airlines:** where to fix operations.
- **Airports:** where and when to add buffer.
- **Managers:** invest in operations or weather response.

Data Sources, Sampling & Data Preparation

Data Source

- **Dataset**: 2015 Flight Delays and Cancellations.
- **Source**: U.S. Department of Transportation – Bureau of Transportation Statistics (BTS).
- **Platform**: Kaggle.
- **Original dataset**: Approximately 5.8 million flights and 31 columns.

Sampling Method

- Random sample representing 0.86% of the original dataset.
- **Final sample: 50,044 flights**



Data Sources, Sampling & Data Preparation

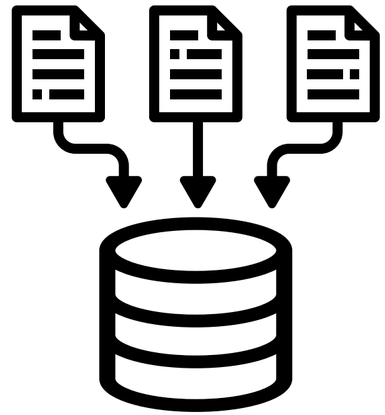
Data Preparation

- Removed records with missing essential information.
- Checked missing values based on flight status.
- Identified incomplete records.
- Handled missing values appropriately.

PostgreSQL Import

Tables created:

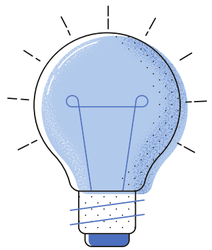
- flights
- flights_raw
- airlines
- airports
- cancellation_reasons



Analytical Data Model / ERD

Objective

Design a relational data model to organize flight information and support the analytical queries.



Data Model

The database consists of four main tables:

- **Flights:** Central table containing flight records, operational information, delays and cancellation status.
- **Airlines:** Stores airline information.
- **Airports:** Contains airport codes, names and geographical information.
- **Cancellation Reasons:** Describes the reasons for flight cancellations.

Key Relationships

- Each flight is associated with an airline.
- Each flight references two airports: origin and destination.
- Each cancelled flight may be associated with a cancellation reason.

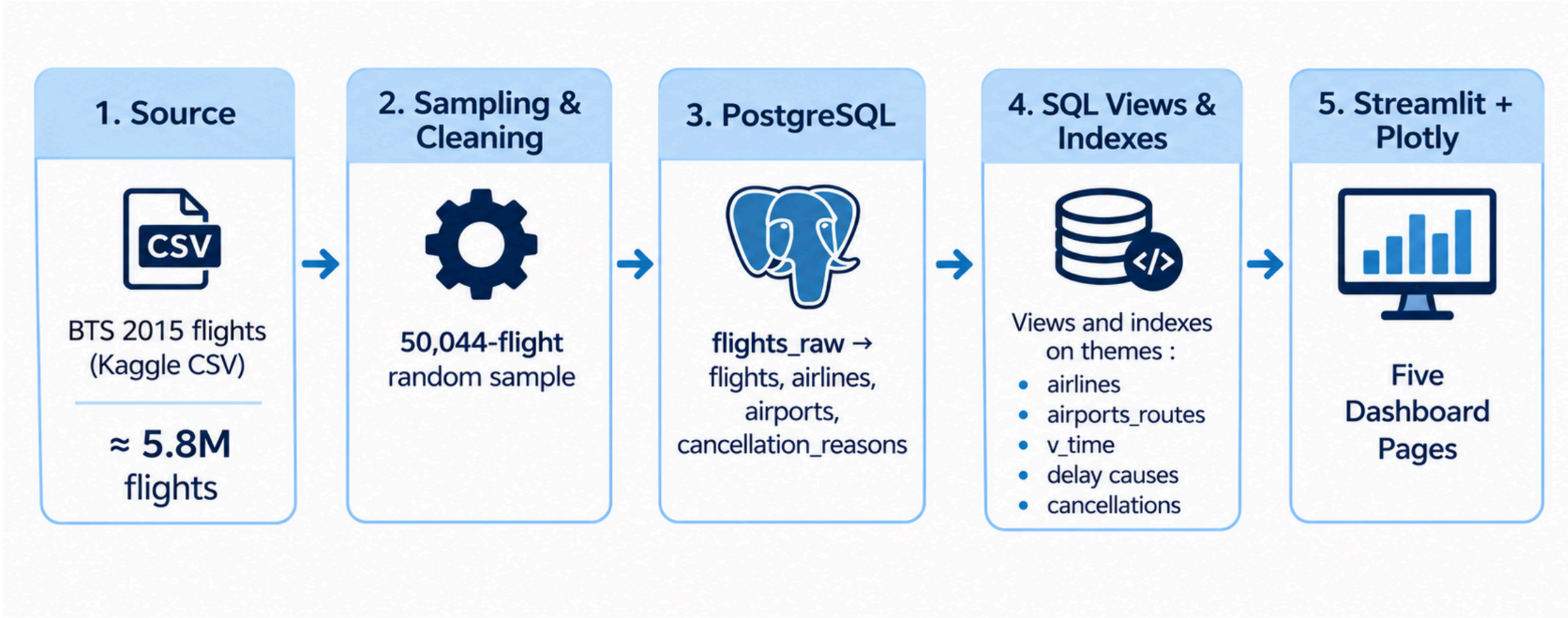
Analytical Data Model / ERD

Entity Relationship Diagram (ERD)

The relational model provides a structured foundation for data preparation, SQL analysis and dashboard development.



Proposed Architecture & Data Flow



Proposed Architecture & Data Flow

CSV (raw data)

the 2015 US flight data from the Bureau of Transportation Statistics, about 5.8 million rows in one flat file. It can’t answer questions on its own.

PostgreSQL (storage)

holds a cleaned sample of 50,044 flights in related tables (flights, airlines, airports, cancellation_reasons), so the whole team works on the same data.

SQL views (rules)

saved queries that count, rate and rank the data, such as the delay rate or the cancellation rate. Each definition is written once, for example delayed means arrival_delay >= 15.

Dashboard (display)

Streamlit and Plotly pages read the views and show charts, KPIs and filters.

Decision-maker (action)

passengers, airlines, airports and managers use the dashboard to decide where to act.

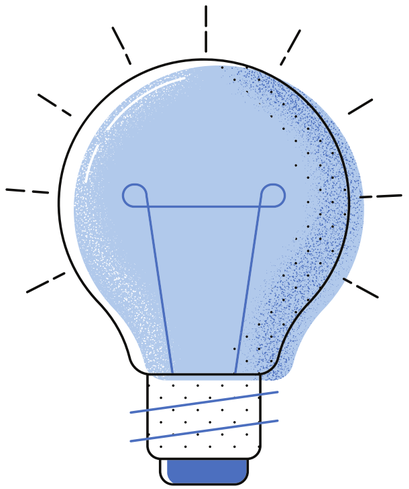
Why PostgreSQL and SQL are important

- **One source of truth**: all five analyses read the same tables, so everyone uses the same numbers.
- **Rules written once:** the delay definition and the 30-flight minimum live in the views and are reused by every dashboard page. If a rule changes, you fix it in one place.
- **Fast, exact counting**: the database groups, counts and ranks rows, and Python only draws the charts. This stays fast if you move from the sample to the full 5.8 million flights.
- **Data quality control**: keys, data types and SQL checks catch NULL delays, invalid codes and unmatched airports. Examples from your project are the 4,165 flights with unmatched airport codes and the NULL delays on cancelled flights.

Airlines: SQL Results & Dashboard Objective

Objective

Identify the airlines with the highest arrival-delay rates.



Delay Definition

A flight is considered delayed when:

ARRIVAL_DELAY >= 15 minutes

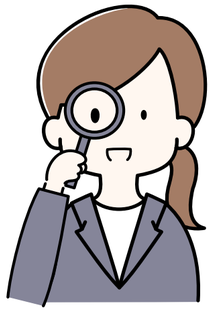
Cancelled flights, diverted flights and flights with missing arrival delay were excluded.

Airlines: SQL Results & Dashboard Objective

Key SQL Results

	airline text	total_flights bigint	delayed_flights bigint	delay_rate numeric
1	Frontier Airlines Inc.	760	211	27.76
2	Spirit Air Lines	980	271	27.65
3	JetBlue Airways	2271	544	23.95
4	American Eagle Airlines Inc.	2326	517	22.23
5	United Air Lines Inc.	4459	946	21.22
6	Atlantic Southeast Airlines	4773	945	19.80
7	US Airways Inc.	1672	328	19.62
8	Southwest Airlines Co.	10533	2009	19.07
9	Virgin America	516	98	18.99
10	American Airlines Inc.	6135	1154	18.81
11	Skywest Airlines Inc.	5089	923	18.14
12	Delta Air Lines Inc.	7528	1029	13.67
13	Alaska Airlines Inc.	1441	175	12.14
14	Hawaiian Airlines Inc.	656	70	10.67

Main Findings

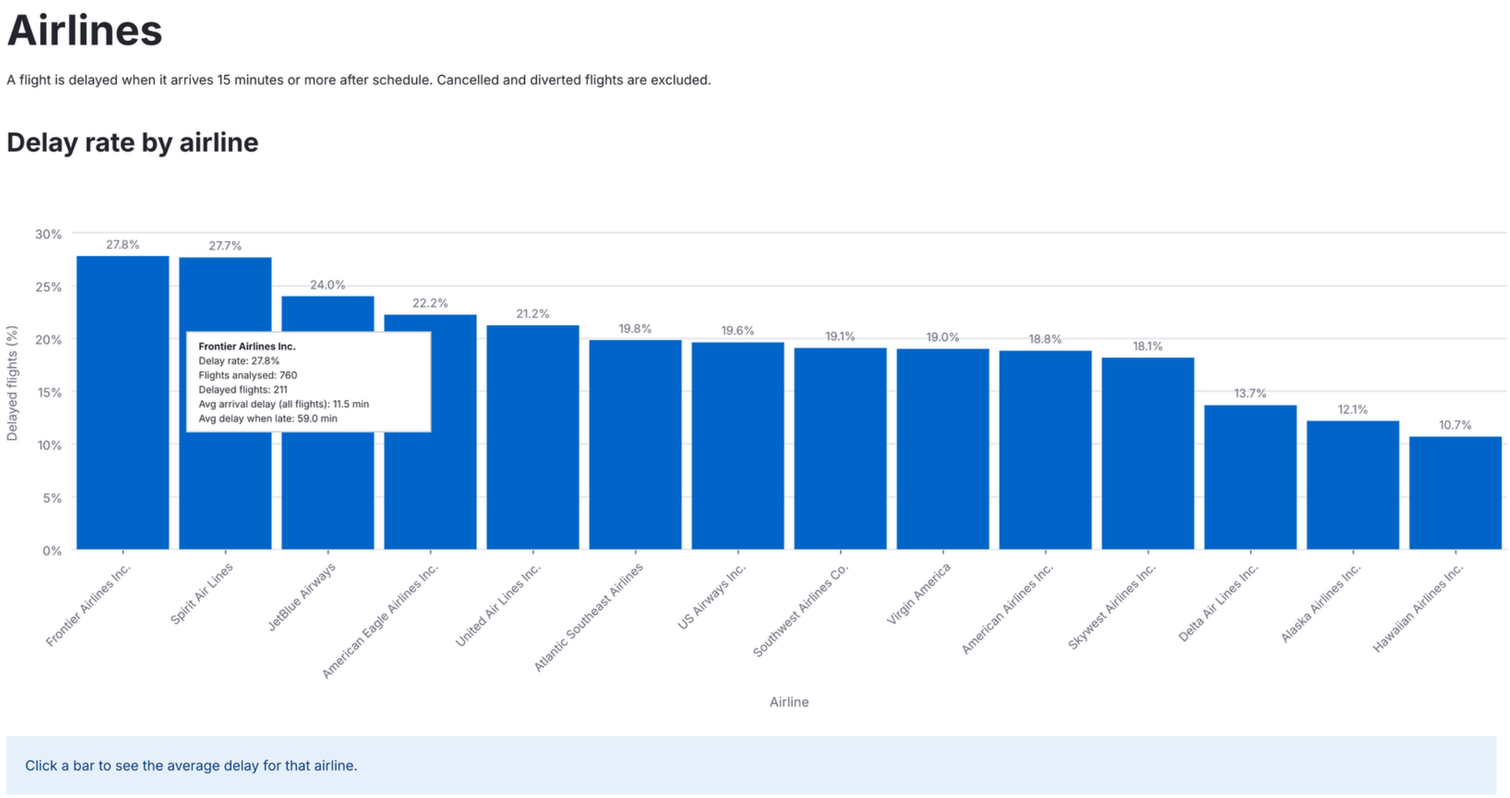


- Frontier and Spirit have the highest observed delay rates.
- Hawaiian has the lowest observed delay rate.
- Delay rates should be interpreted alongside the number of flights.

Airlines: SQL Results & Dashboard Objective

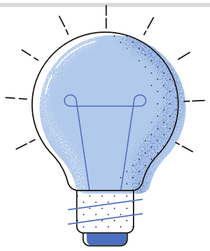
Dashboard

Interactive dashboard developed with Python, Pandas, Plotly and Streamlit.



Airports & Routes: SQL Results & Dashboard

Objective



Identify the airports and routes most affected by arrival delays.

Key SQL Results

1. Top Airports - Delay Volume

	airport_code text	airport_name text	city text	state text	eligible_flights bigint	delayed_flights bigint	delay_rate_pct numeric	avg_arrival_delay numeric	delay_volume_rank bigint	delay_rate_rank bigint
1	ORD	Chicago O'Hare International Airport	Chicago	IL	2376	592	24.92	62.95	1	7
2	ATL	Hartsfield-Jackson Atlanta International Airp…	Atlanta	GA	2926	466	15.93	55.50	2	96
3	DFW	Dallas/Fort Worth International Airport	Dallas-Fort Wor…	TX	1983	428	21.58	57.22	3	28
4	DEN	Denver International Airport	Denver	CO	1664	397	23.86	53.40	4	13
5	LAX	Los Angeles International Airport	Los Angeles	CA	1655	350	21.15	59.45	5	35
6	IAH	George Bush Intercontinental Airport	Houston	TX	1262	266	21.08	58.94	6	38
7	SFO	San Francisco International Airport	San Francisco	CA	1270	254	20.00	59.28	7	47
8	PHX	Phoenix Sky Harbor International Airport	Phoenix	AZ	1218	234	19.21	43.81	8	52
9	LGA	LaGuardia Airport (Marine Air Terminal)	New York	NY	847	224	26.45	60.63	9	6
10	LAS	McCarran International Airport	Las Vegas	NV	1196	222	18.56	50.80	10	60

Airports & Routes: SQL Results & Dashboard

Key SQL Results

2. Top Airports - Delay Rate

	airport_code text	airport_name text	city text	state text	eligible_flights bigint	delayed_flights bigint	delay_rate_pct numeric	avg_arrival_delay numeric	delay_volume_rank bigint	delay_rate_rank bigint
1	ACY	Atlantic City International Airport	Atlantic City	NJ	35	13	37.14	84.85	83	1
2	LBB	Lubbock Preston Smith International Airport	Lubbock	TX	37	12	32.43	66.92	85	2
3	ASE	Aspen-Pitkin County Airport	Aspen	CO	32	9	28.13	55.00	101	3
4	DAY	James M. Cox Dayton International Airport	Dayton	OH	78	21	26.92	45.67	61	4
5	STT	Cyril E. King Airport	Charlotte Ama…	VI	41	11	26.83	185.91	89	5
6	LGA	LaGuardia Airport (Marine Air Terminal)	New York	NY	847	224	26.45	60.63	9	6
7	ORD	Chicago O'Hare International Airport	Chicago	IL	2376	592	24.92	62.95	1	7
8	SBA	Santa Barbara Municipal AirportÂ (Santa Barbara Airp…	Santa Barbara	CA	61	15	24.59	41.53	74	8
9	CHS	Charleston International Airport/Charleston AFB	Charleston	SC	106	26	24.53	84.54	56	9
10	LFT	Lafayette Regional Airport	Lafayette	LA	41	10	24.39	49.20	94	10

Airports & Routes: SQL Results & Dashboard

Key SQL Results

2. Top Routes - Delay Volume

	origin_code text	origin_name text	origin_city text	destination_code text	destination_name text	destination_city text	eligible_flights bigint	delayed_flights bigint	delay_rate_pct numeric
1	SFO	San Francisco International Airport	San Francisco	LAX	Los Angeles International Airport	Los Angeles	120	31	25.83
2	ORD	Chicago O'Hare International Airport	Chicago	LGA	LaGuardia Airport (Marine Air Terminal)	New York	72	29	40.28
3	LAX	Los Angeles International Airport	Los Angeles	SFO	San Francisco International Airport	San Francisco	116	25	21.55
4	LAX	Los Angeles International Airport	Los Angeles	JFK	John F. Kennedy International AirportÂ (New York International Airp…	New York	108	24	22.22
5	SAN	San Diego International AirportÂ (Lindbergh Fi…	San Diego	SFO	San Francisco International Airport	San Francisco	61	23	37.70
6	ATL	Hartsfield-Jackson Atlanta International Airport	Atlanta	LGA	LaGuardia Airport (Marine Air Terminal)	New York	72	23	31.94
7	LAS	McCarran International Airport	Las Vegas	LAX	Los Angeles International Airport	Los Angeles	80	23	28.75
8	LAX	Los Angeles International Airport	Los Angeles	LAS	McCarran International Airport	Las Vegas	86	23	26.74
9	DFW	Dallas/Fort Worth International Airport	Dallas-Fort Wor…	DEN	Denver International Airport	Denver	54	21	38.89
10	LAS	McCarran International Airport	Las Vegas	SFO	San Francisco International Airport	San Francisco	73	21	28.77

Airports & Routes: SQL Results & Dashboard

Dashboard

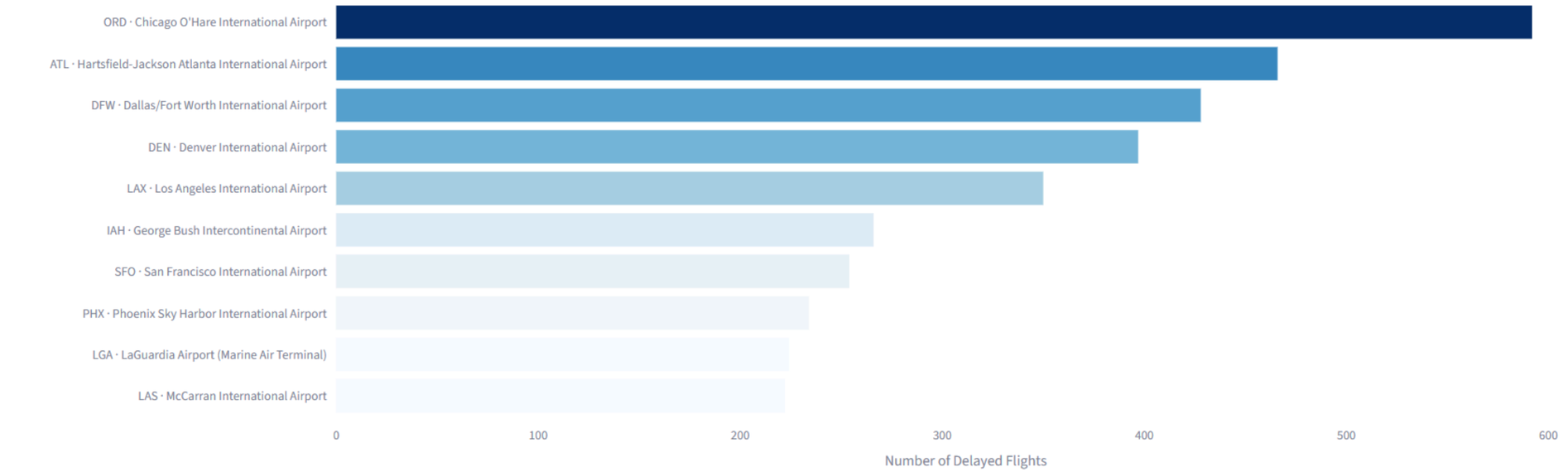
01. Airport Delay Analysis

Explore airports with the highest number and percentage of delayed flights.

Rank airports by: Delayed Flights

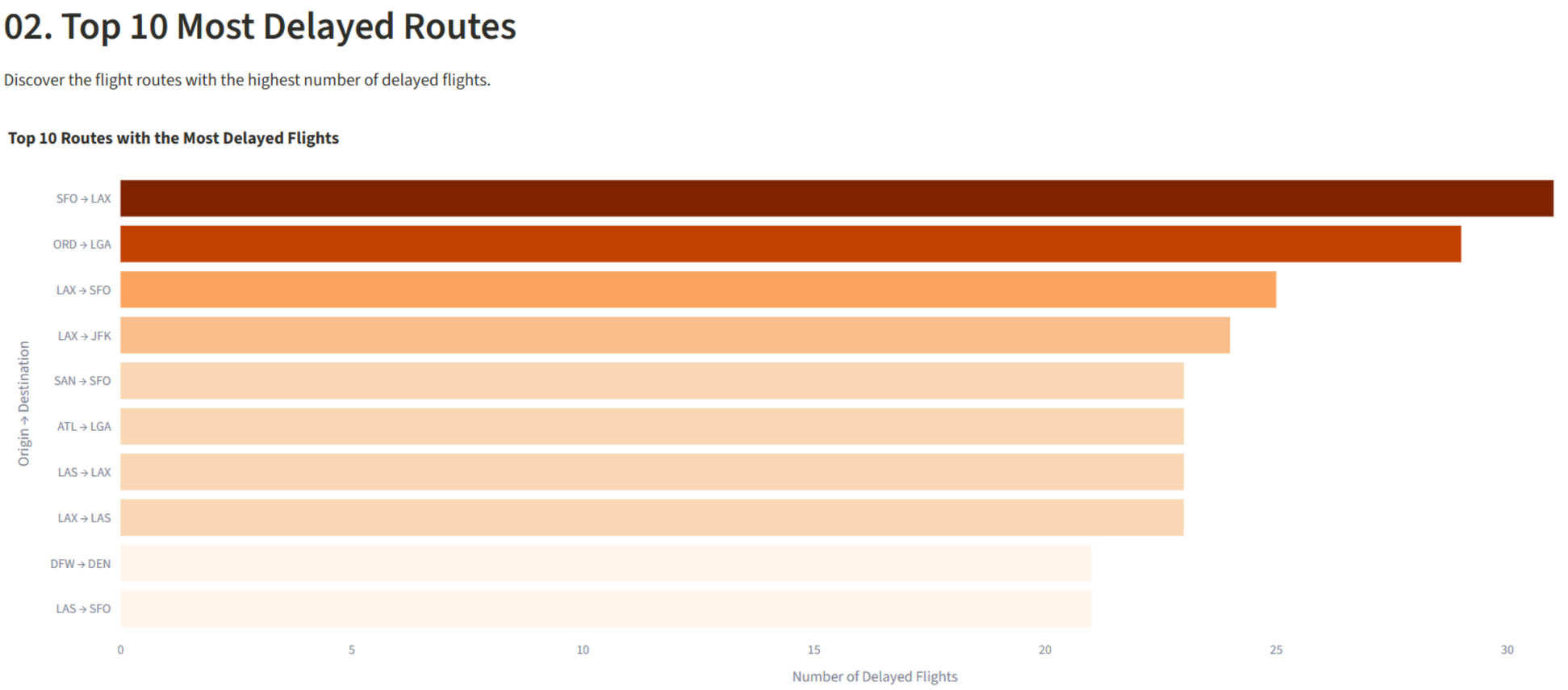
Number of airports: 10

Airports with the Highest Delay Volume



Airports & Routes: SQL Results & Dashboard

Dashboard



Airports & Routes: SQL Results & Dashboard

Dashboard

Airports & Routes Analysis

Flight Delays Analysis | 2015

Key Performance Indicators

Most Delayed Airport	Highest Airport Delay Rate	Most Delayed Route	Highest Route Delay Rate (Top 10)
ORD	37.14%	SFO → LAX	40.28%
↑ 592 delayed flights	↑ ACY	↑ 31 delayed flights	↑ ORD → LGA

Main Findings

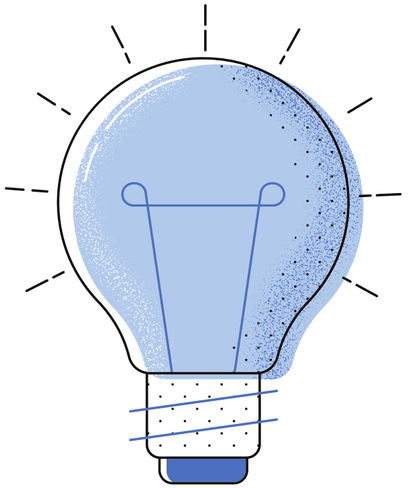
- **ORD** records the highest number of delayed flights (**592**).
- **ACY** has the highest observed delay rate (**37.14%**).
- **SFO–LAX** is the route with the highest delayed-flight volume (**31**).
- Delay volume and delay rate reveal different aspects of airport performance.



Time Patterns: SQL Results & Dashboard

Objective

- Identify when delays happen most often: by hour, weekday, month and time of day.



Delay definition: ARRIVAL_DELAY >= 15 minutes. Cancelled flights, diverted flights and flights with a missing arrival delay are excluded.

The hour is the scheduled departure hour. The delay is measured at arrival.

Time Patterns: SQL Results & Dashboard

Key SQL Results

1. Delay Patterns by Departure Hour and Weekday

Identifies combinations of departure hours and weekdays with the highest delay rates, revealing when delays are most concentrated.

	day_of_week smallint	day_name text	hour_of_day smallint	nb_flights integer	delay_rate_pct numeric
1	6	Saturday	23	40	32.50
2	1	Monday	22	169	31.95
3	4	Thursday	21	241	31.54
4	7	Sunday	21	273	31.50
5	1	Monday	18	390	30.77
6	1	Monday	21	220	30.45
7	4	Thursday	18	418	29.90
8	4	Thursday	19	426	29.58
9	5	Friday	18	426	28.87
10	4	Thursday	17	511	28.57

Time Patterns: SQL Results & Dashboard

Key SQL Results

3.Delay Rates by Departure Hour

Identifies the hours with the highest and lowest delay rates, helping reveal whether departure time influences flight punctuality.

	r_of_day illint	nb_flights integer	nb_delayed integer	delay_rate_pct numeric	ci_low_pct numeric	ci_high_pct numeric	avg_delay_when_late_min numeric	vs_overall_pts numeric	signal text	worst_rank bigint
1	18	2768	744	26.88	25.26	28.56	63.8	8.12	Above over…	1
2	19	2796	731	26.14	24.55	27.81	59.2	7.38	Above over…	2
3	21	1655	422	25.50	23.46	27.65	57.6	6.74	Above over…	3
4	17	3302	826	25.02	23.57	26.52	58.5	6.26	Above over…	4
5	20	2215	545	24.60	22.86	26.44	58.8	5.84	Above over…	5
6	22	1039	249	23.97	21.47	26.65	56.1	5.21	Above over…	6
7	16	2866	673	23.48	21.97	25.07	60.3	4.72	Above over…	7
8	15	3075	700	22.76	21.32	24.28	58.4	4.00	Above over…	8
9	23	400	87	21.75	17.99	26.05	45.2	2.99	In line	9
10	13	3016	601	19.93	18.54	21.39	56.2	1.17	In line	10
11	14	2734	544	19.90	18.44	21.44	56.6	1.14	In line	11

Time Patterns: SQL Results & Dashboard

Key SQL Results

3. Monthly Flight Delay Trends

Tracks monthly delay rates to identify seasonal patterns and changes in flight punctuality over time.

	year smallint	month smallint	period text	month_name text	nb_flights integer	nb_delayed integer	delay_rate_pct numeric	ci_low_pct numeric	ci_high_pct numeric	prev_month_rate_pct numeric	change_pts numeric
1	2015	1	2015-01	Jan	3939	822	20.87	19.63	22.17	[null]	[r
2	2015	2	2015-02	Feb	3476	790	22.73	21.36	24.15	20.87	1
3	2015	3	2015-03	Mar	4227	801	18.95	17.80	20.16	22.73	-3
4	2015	4	2015-04	Apr	4116	699	16.98	15.87	18.16	18.95	-1
5	2015	5	2015-05	May	4209	784	18.63	17.48	19.83	16.98	1
6	2015	6	2015-06	Jun	4234	983	23.22	21.97	24.51	18.63	4
7	2015	7	2015-07	Jul	4431	1000	22.57	21.36	23.82	23.22	-0
8	2015	8	2015-08	Aug	4335	839	19.35	18.21	20.56	22.57	-3
9	2015	9	2015-09	Sep	3971	498	12.54	11.55	13.61	19.35	-6
10	2015	10	2015-10	Oct	4140	556	13.43	12.43	14.50	12.54	0
11	2015	11	2015-11	Nov	3982	599	15.04	13.97	16.19	13.43	1

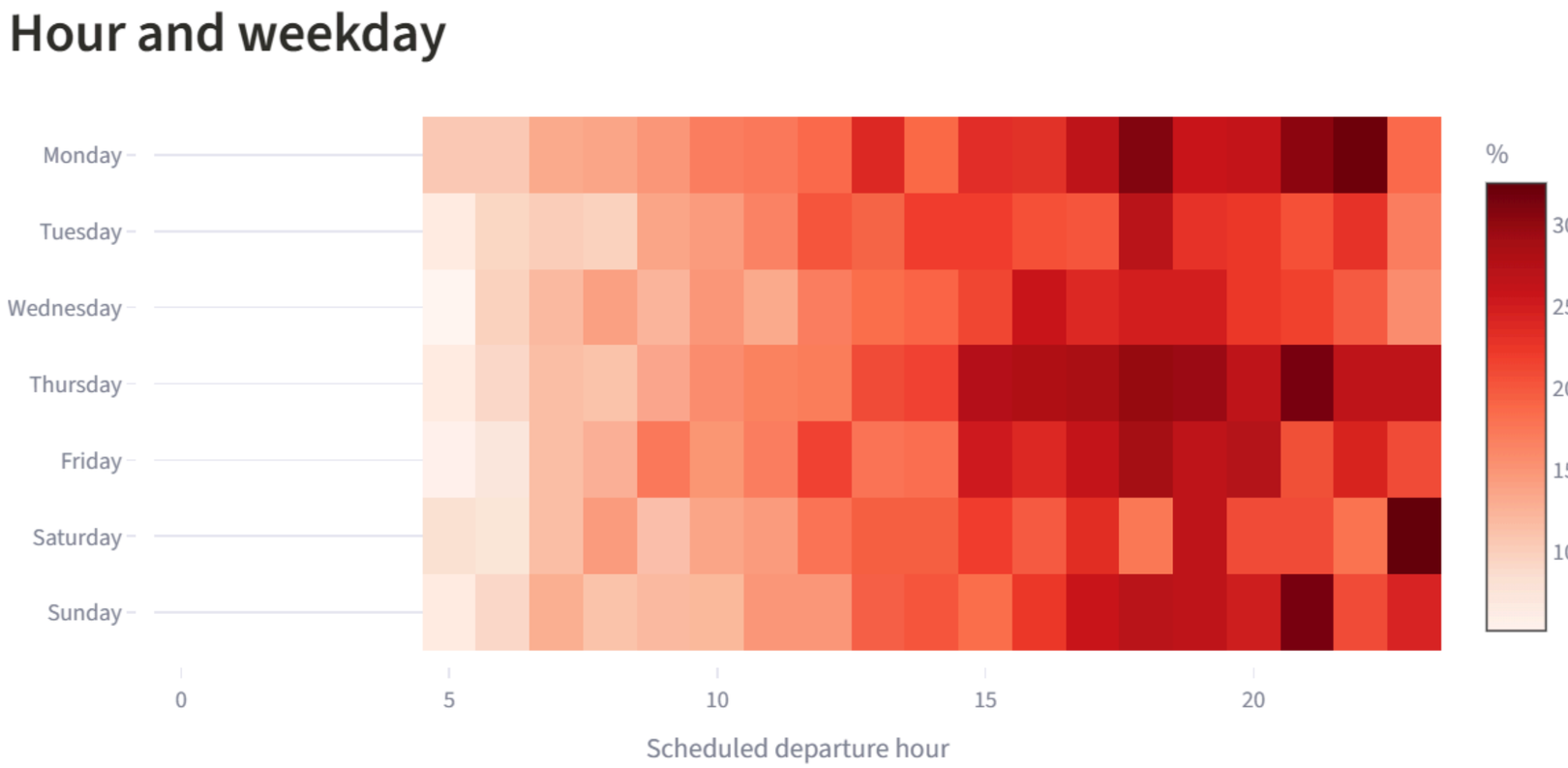
Time Patterns: SQL Results & Dashboard

Dashboard

1.Delay patterns by hour and weekday (heatmap)

Show the interactive heatmap of delay rates by scheduled departure hour and weekday. Explain:

- Which combinations have the highest delay rates?
- Are delays concentrated at particular hours?
- Are some weekdays more affected?

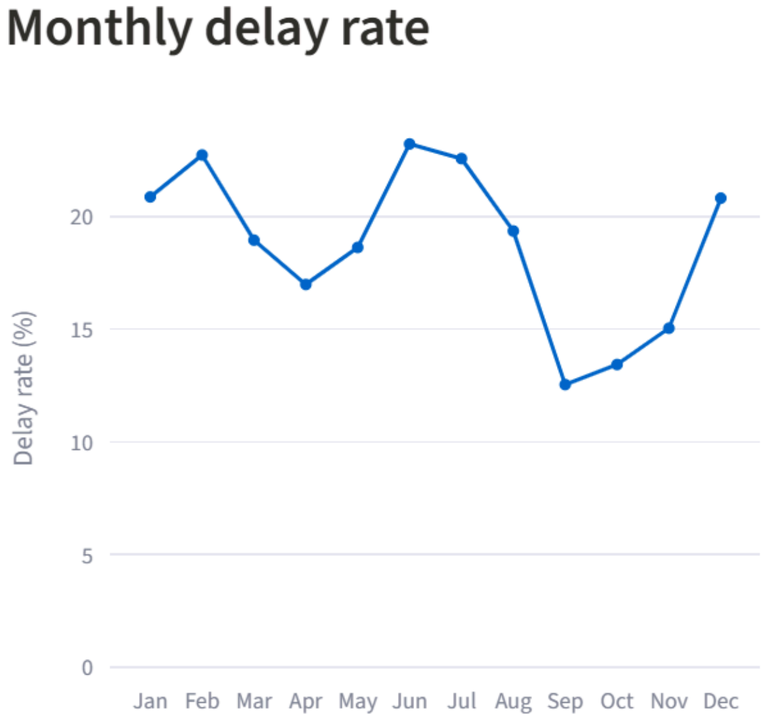


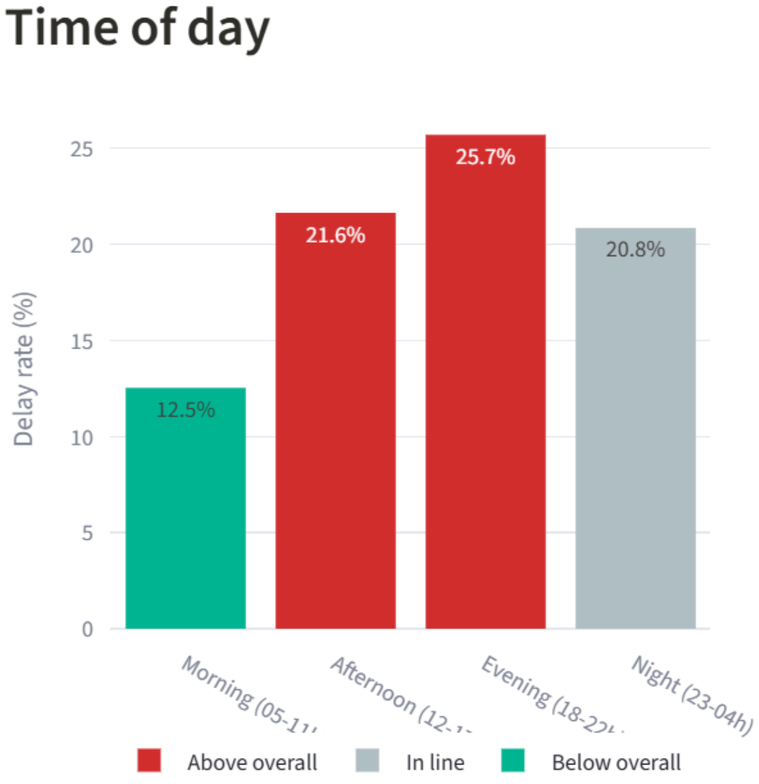
Time Patterns: SQL Results & Dashboard

Dashboard

2.Monthly delay trends

Show how the delay rate changes over the months.
Explain whether delays increase, decrease or fluctuate over time.





Time Patterns: SQL Results & Dashboard

Dashboard

When do delays happen?

A flight is delayed when it arrives 15 minutes or more after schedule. Cancelled and diverted flights are excluded. Hours are scheduled departure hours.

Delay rate	Worst hour	Worst weekday	Worst period
18.76 %	18:00	Thursday	Evening
↑ 9,220 of 49,139 flights	↑ 26.88 %	↑ 20.40 %	(18-22h) ▲ 25.69 %

Main Findings

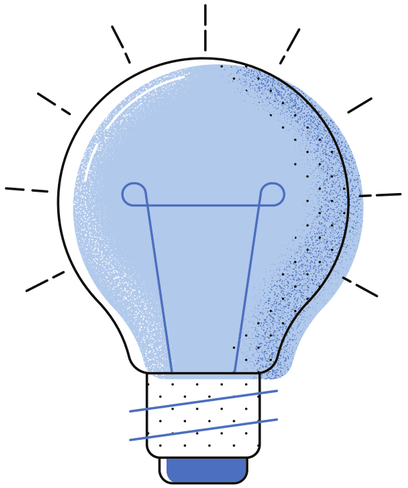
- Delays peak at [18:00] (26.88%) .
- The Evening period (18h-22h) is the worst time of day (25.69%).
- Thursday is the worst weekday (20.40%).



Delay Causes : SQL Results & Dashboard

Objective

- Find out what is the main cause driving the delays



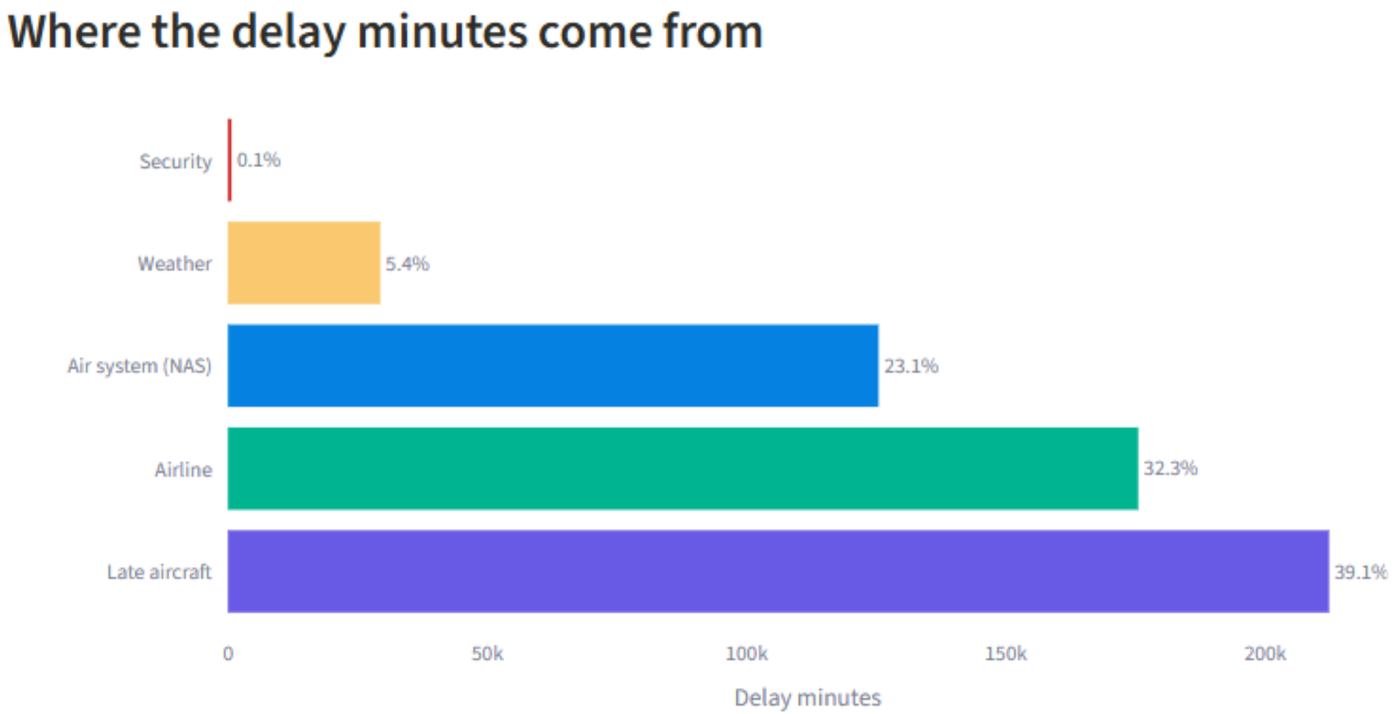
which cause contributed most to delays across the different airports ,
causes range from : airline delays , late aircrafts , security procedures , weather problems etc ...

Delay Causes : SQL Results & Dashboard

Dashboard

Two causes explain 71% of delay minutes

Late aircraft and airline controlled causes together produce most of the lost minutes.

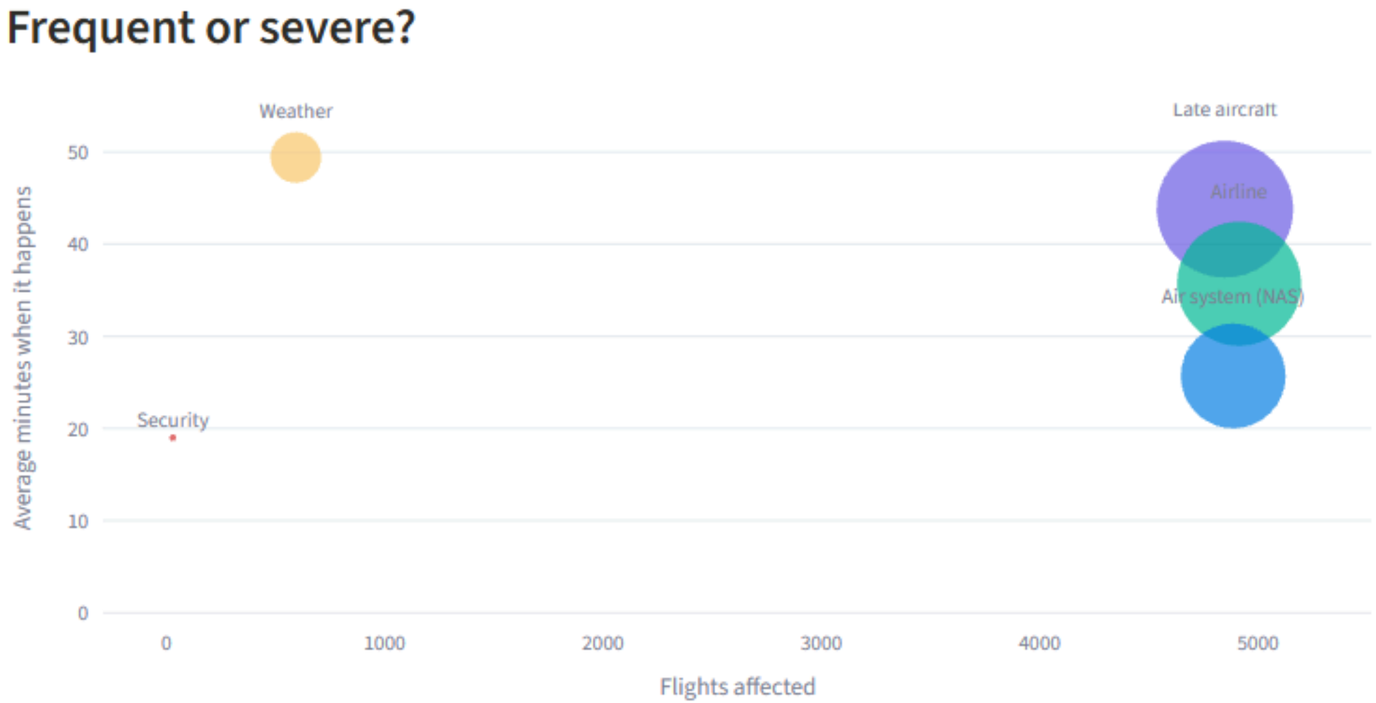


Delay Causes : SQL Results & Dashboard

Dashboard

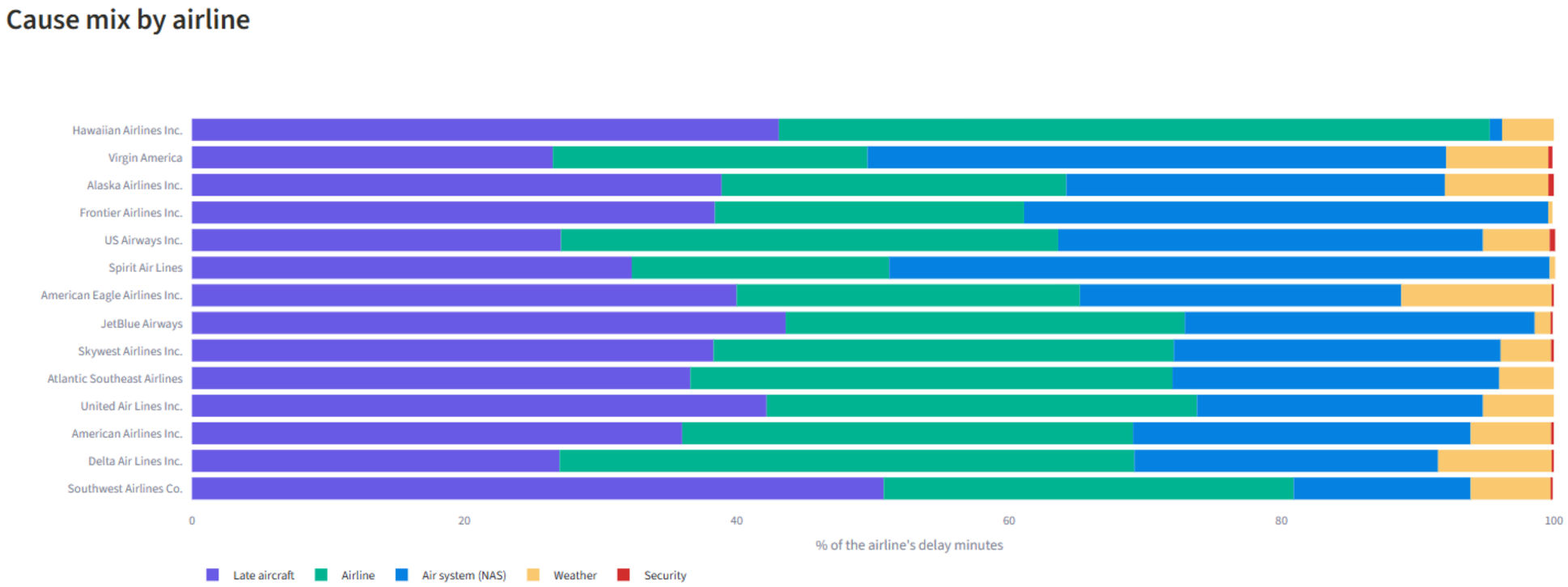
is it due to frequency rather than magnitude ?

the bubble size corresponds to the total minutes per delay cause



Delay Causes : SQL Results & Dashboard

Dashboard



notice the consistency here, all of them experienced delays due to late aircrafts, airline delays and airsystem delays

Delay Causes : SQL Results & Dashboard

Main Findings

- 71% of delay minutes come from two causes: late aircraft (39%) and airline-controlled causes (32%)
- Weather is rare but severe: 594 flights (5% of minutes), yet 49 minutes per flight, the longest of any cause. Security is negligible (30 flights, 0.1%)
- Each airline has its own profile: Southwest 51% late aircraft, Delta 42% airline-controlled, Spirit 49% air system

Delay Causes : SQL Results & Dashboard

What to do about it ?

- **71% of delay minutes come from two causes: late aircraft (39%) and airline-controlled causes (32%)**

→ buffer time

- **Weather is rare but severe : 5% of minutes yet 49 minutes per flight**

→ prepare a contingency plan for the rare large events

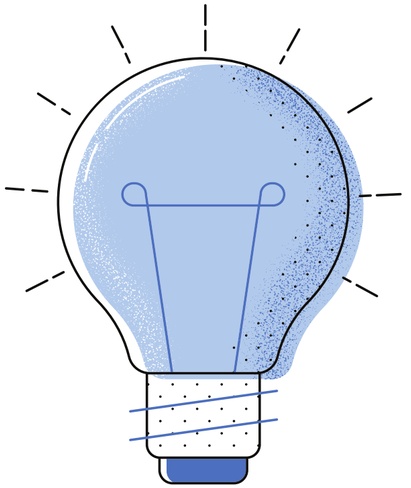
- **Each airline has its own profile: Southwest 51% late aircraft, Delta 42% airline-controlled, Spirit 49% air system**

→ Use one fix per profile

Cancellations: SQL Results & Dashboard

Objective

- Understand when and why flights are cancelled.



Cancellation rate = cancelled flights / all scheduled flights. Cancelled flights stay in the denominator.

A CASE expression turns the codes A, B, C, D into Airline, Weather, Air system (NAS) and Security.

Cancellations: SQL Results & Dashboard

Key SQL Results

1 Main Reasons for Flight Cancellations

Identifies the leading causes of cancellations and measures each reason's contribution to total cancelled flights.

	reason_code text	reason text	nb_cancelled integer	share_pct numeric	per_1000_flights numeric
1	B	Weather	424	54.50	8.47
2	A	Airline	210	26.99	4.20
3	C	Air system (NA…	144	18.51	2.88

Cancellations: SQL Results & Dashboard

Key SQL Results

2. Cancellation Rates by Airline

Compares cancellation rates across airlines to identify which airlines experience higher cancellation rates.

	airline_code text	airline_name text	nb_flights integer	nb_cancelled integer	cancel_rate_pct numeric	ci_low_pct numeric	ci_high_pct numeric	vs_overall_pts numeric	signal text	nb_airlir integer
1	MQ	American Eagle Airlines I...	2471	137	5.54	4.71	6.52	3.99	Above over...	
2	EV	Atlantic Southeast Airlines	4917	131	2.66	2.25	3.15	1.11	Above over...	
3	B6	JetBlue Airways	2316	44	1.90	1.42	2.54	0.35	In line	
4	US	US Airways Inc.	1708	30	1.76	1.23	2.50	0.21	In line	
5	OO	Skywest Airlines Inc.	5188	85	1.64	1.33	2.02	0.09	In line	
6	NK	Spirit Air Lines	995	15	1.51	0.92	2.47	-0.04	In line	
7	AA	American Airlines Inc.	6235	89	1.43	1.16	1.75	-0.12	In line	
8	UA	United Air Lines Inc.	4537	64	1.41	1.11	1.80	-0.14	In line	
9	WN	Southwest Airlines Co.	10701	136	1.27	1.08	1.50	-0.28	Below over...	
10	F9	Frontier Airlines Inc.	766	5	0.65	0.28	1.52	-0.90	Below over...	
11	AS	Alaska Airlines Inc.	1455	9	0.62	0.33	1.17	-0.93	Below over...	

Cancellations: SQL Results & Dashboard

Key SQL Results

3. Cancellation Reasons Across Months

hows how cancellation causes change over time, revealing seasonal patterns and periods when specific reasons become more frequent.

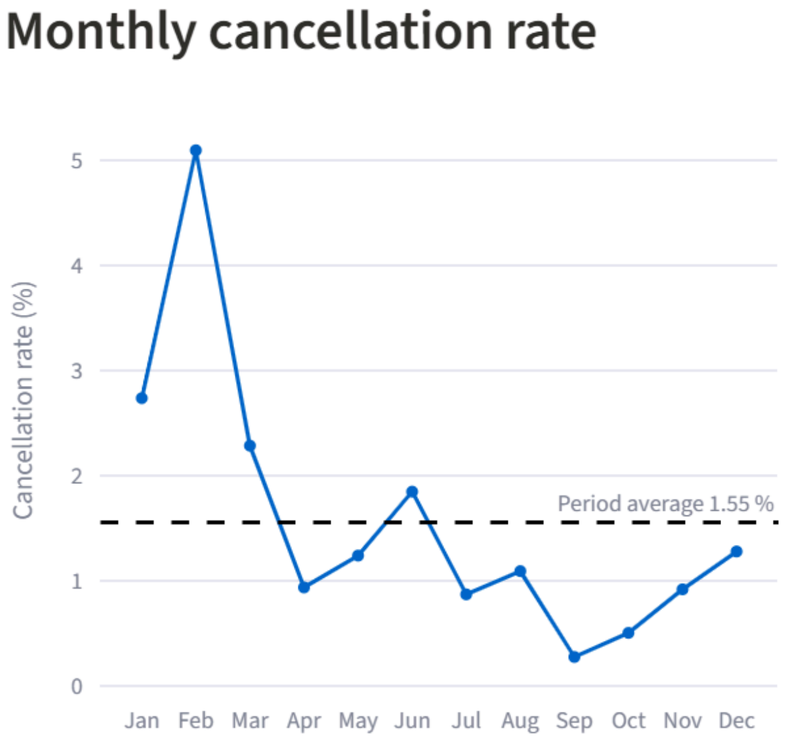
	year smallint	month smallint	period text	month_name text	reason_code text	reason text	nb_cancelled integer	nb_flights integer	per_1000_flights numeric	share_of_month_pct numeric
1	2015	12	2015-12	Dec	A	Airline	19	4146	4.58	35.8
2	2015	8	2015-08	Aug	A	Airline	21	4397	4.78	43.8
3	2015	4	2015-04	Apr	A	Airline	16	4164	3.84	41.0
4	2015	7	2015-07	Jul	A	Airline	24	4480	5.36	61.5
5	2015	9	2015-09	Sep	A	Airline	6	3993	1.50	54.5
6	2015	11	2015-11	Nov	A	Airline	6	4028	1.49	16.2
7	2015	2	2015-02	Feb	A	Airline	30	3672	8.17	16.0
8	2015	6	2015-06	Jun	A	Airline	21	4331	4.85	26.3
9	2015	10	2015-10	Oct	A	Airline	11	4165	2.64	52.4
10	2015	5	2015-05	May	A	Airline	11	4277	2.57	20.8
11	2015	1	2015-01	Jan	A	Airline	19	4057	4.68	17.1

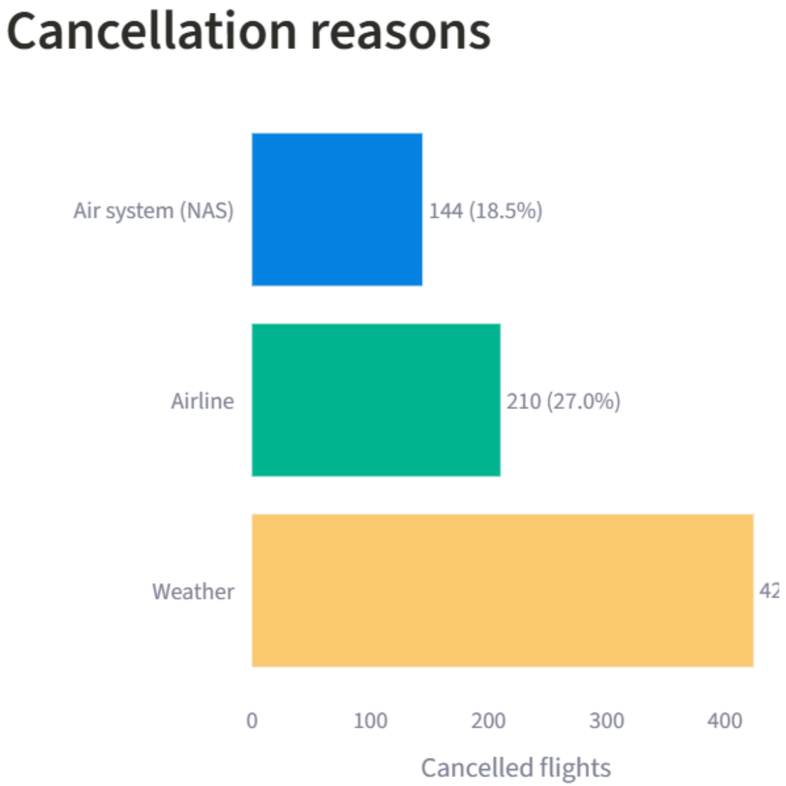
Cancellations: SQL Results & Dashboard

Dashboard

1.Cancellation reasons

Explain which causes account for the largest share of cancellations.





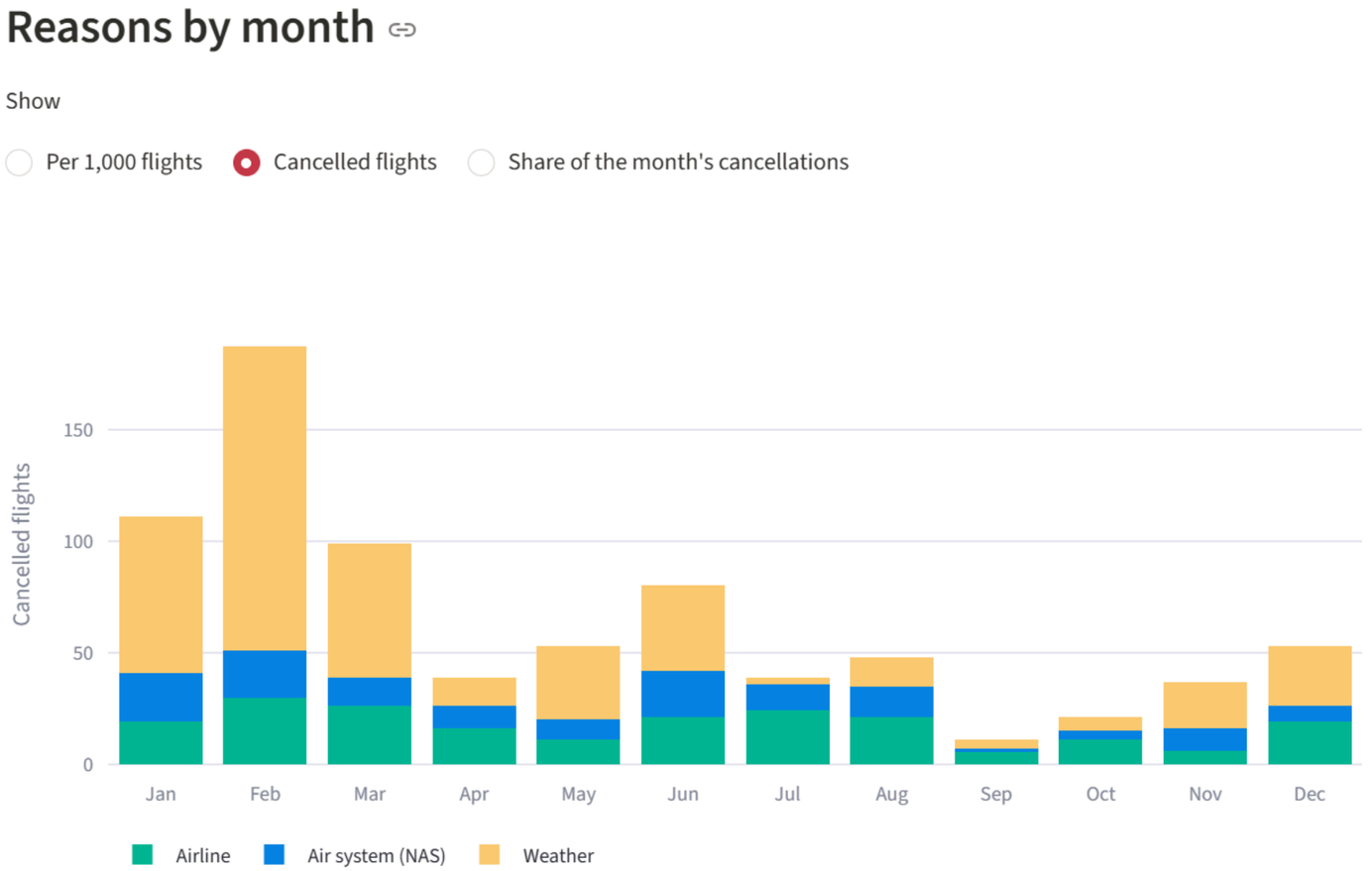
Cancellations: SQL Results & Dashboard

Dashboard

2.Cancellation trends by month

Show the monthly cancellation-rate line chart.
Explain:

- Which months have the highest cancellation rates?
- Are cancellations seasonal or irregular?
- When should the airline prepare additional operational capacity?



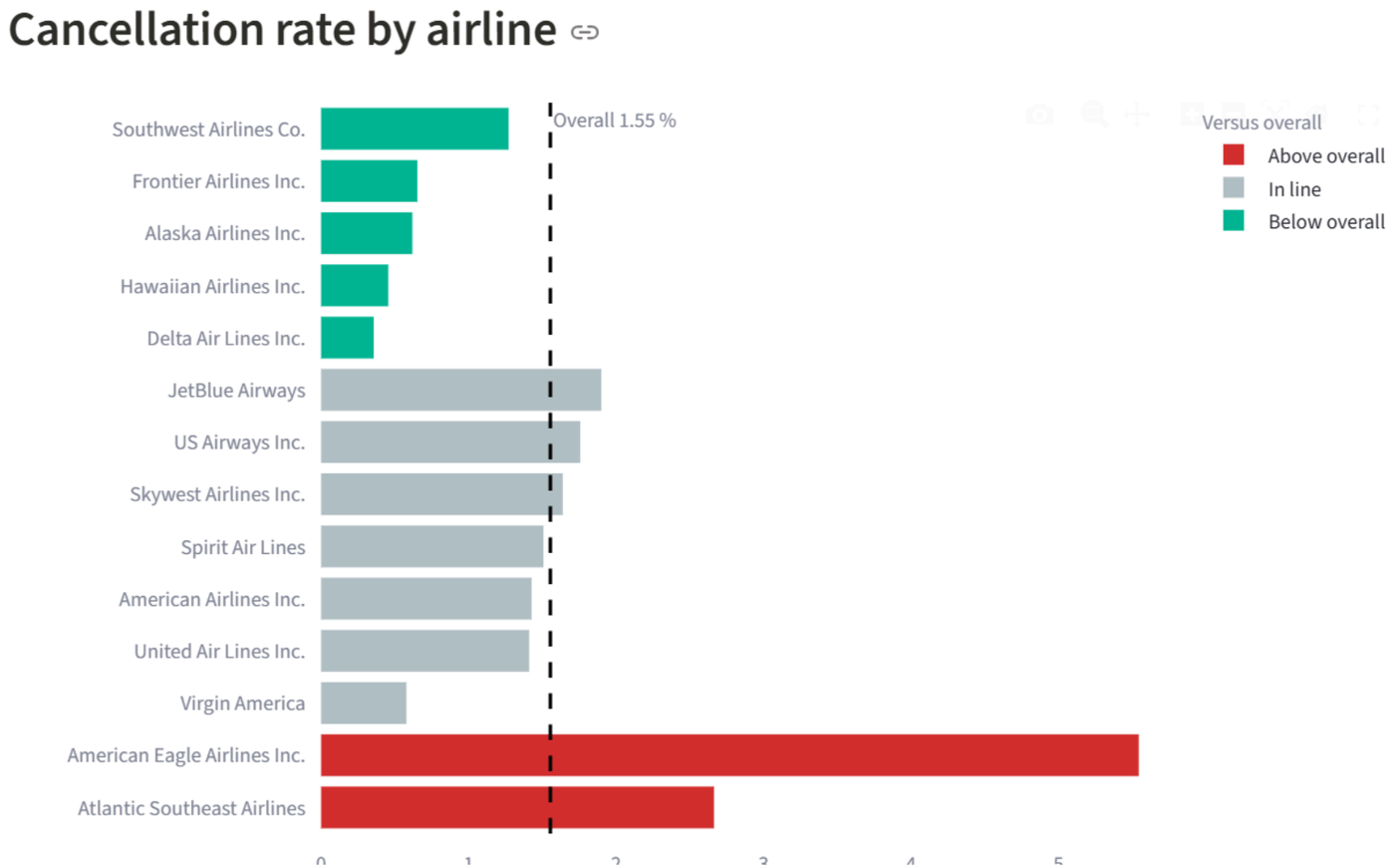
Cancellations: SQL Results & Dashboard

Dashboard

3.Cancellation rates by airline

Show the airline comparison chart.
Explain:

- Which airlines have higher cancellation rates?
- Which airlines are performing better?
- Which airlines are clearly above the overall cancellation rate?



Cancellations: SQL Results & Dashboard

Dashboard

When and why are flights cancelled?

Cancellation rate = cancelled flights / all scheduled flights. Cancelled flights have no delay values, so they are never used in delay averages.

Cancellation rate	Worst month	Main reason	Airline-caused
1.55 %	Feb	Weather	27.0 %
↑ 778 of 50,044 flights	↑ 5.09 %	↑ 54.5 % of cancellations	↑ of cancellations

Main Findings

- 788 of 50,044 flights were cancelled (1.55%).
- The worst month is february (5.09%).
- [Weather] causes 54.5 % of cancellations, and 27 % are airline-caused, the part airlines can control.



Three Key Findings & Recommendations

1. Delay causes

About 71% of delay minutes come from late aircraft (39.1%) and airline-controlled causes (32.3%).

Recommendation

Add buffer time to tight aircraft rotations.

2. Delay risk

Delay risk is concentrated: Frontier (27.76%) and Spirit (27.65%) against Hawaiian (10.67%); ORD has the most delayed flights (592); delays peak at 18:00 (26.88%).

Recommendation

Add buffers at the busiest airports and in the evening, the worst time of day (25.69%).

3. Cancellations

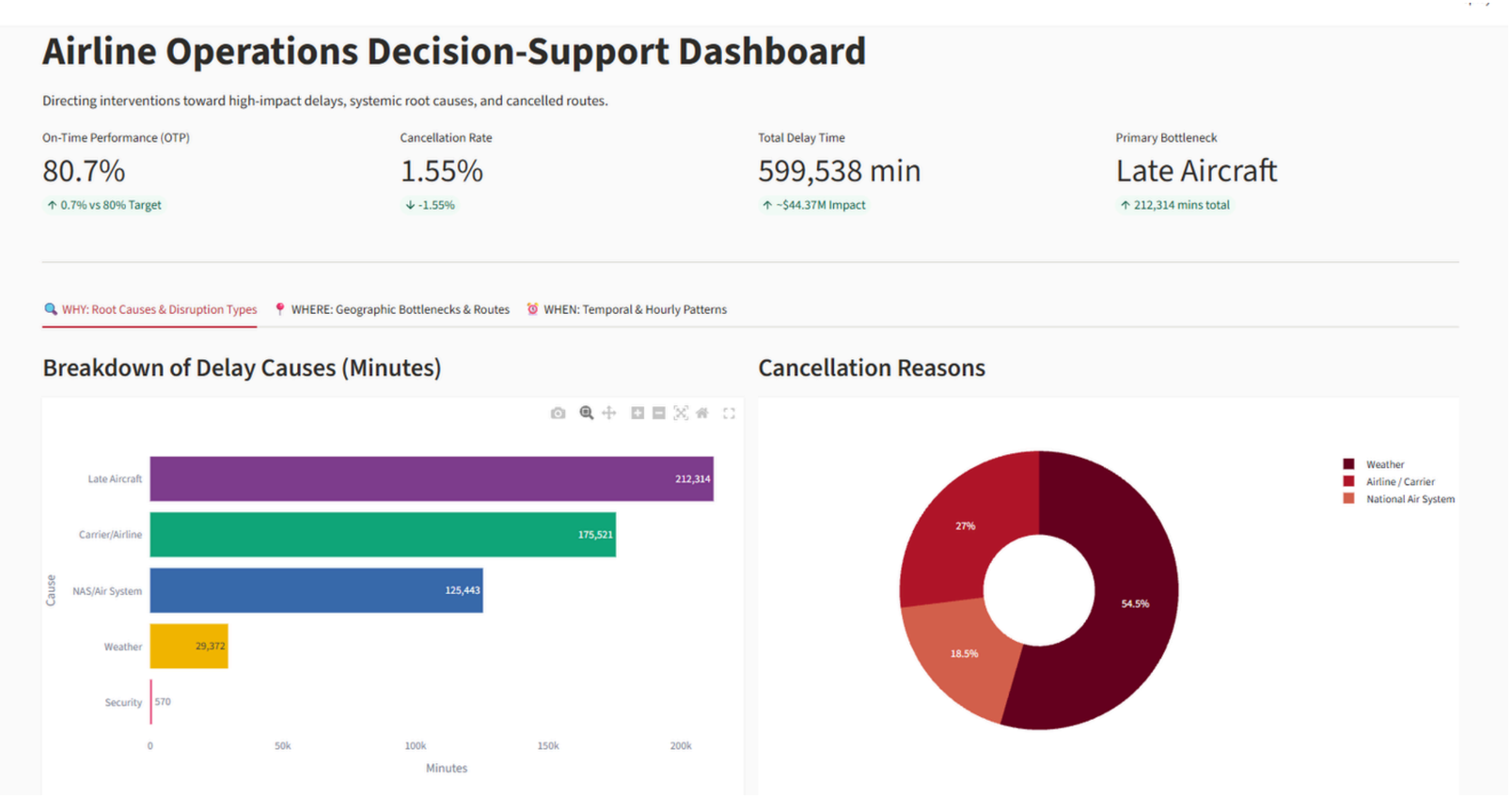
Weather causes 54.5% of cancellations, peaking in February (5.09% against a 1.55% average).

Recommendation

Weather: cancel and rebook earlier and plan winter capacity. Airline-caused (27%): crew, aircraft and maintenance planning.

AI Comparison

Gemini’s dashboard and number mismatch



AI Comparison

Results Comparison

	Gemini	Ours
On-time rate	80.70%	81.20%
Total delay minutes	599,538	543,220
Cost of delay	$44M (invented)	none, no source
Top 10 routes	led by routes with 1 flight	minimum 30 flights

AI Comparison

Results Comparison

What the AI did well

- Clear structure: why, where, when
- Filters by airline and month
- Its delay cause totals match ours exactly



What we had to check

- Two definitions of delay in the same page
- Invented cost and target
- Calculations done in Python on every row, not in SQL
- Airports with numeric codes disappear without a warning



Open Issues & Proposed Improvement

Open Issue — Unmatched Airport Codes

- **4,165 flights (8.32%)** have airport codes that could not be matched with the airport reference table.
- This may affect the **completeness** of airport and route analyses.



Proposed Improvement

- Standardize airport codes across datasets.
- Investigate unmatched codes and improve airport data mapping.

Open Issues & Proposed Improvement

Open Issue — Reliability

Some airlines have fewer flights, which may make their delay rates less reliable.

Reliability Analysis

- Calculated the margin of error to assess result reliability.
- Used 1,000 flights as a minimum threshold.



Conclusion

- Most airlines have enough observations for a more reliable comparison.
- Results for airlines with fewer than 1,000 flights should be interpreted with caution.

Thank **You**

